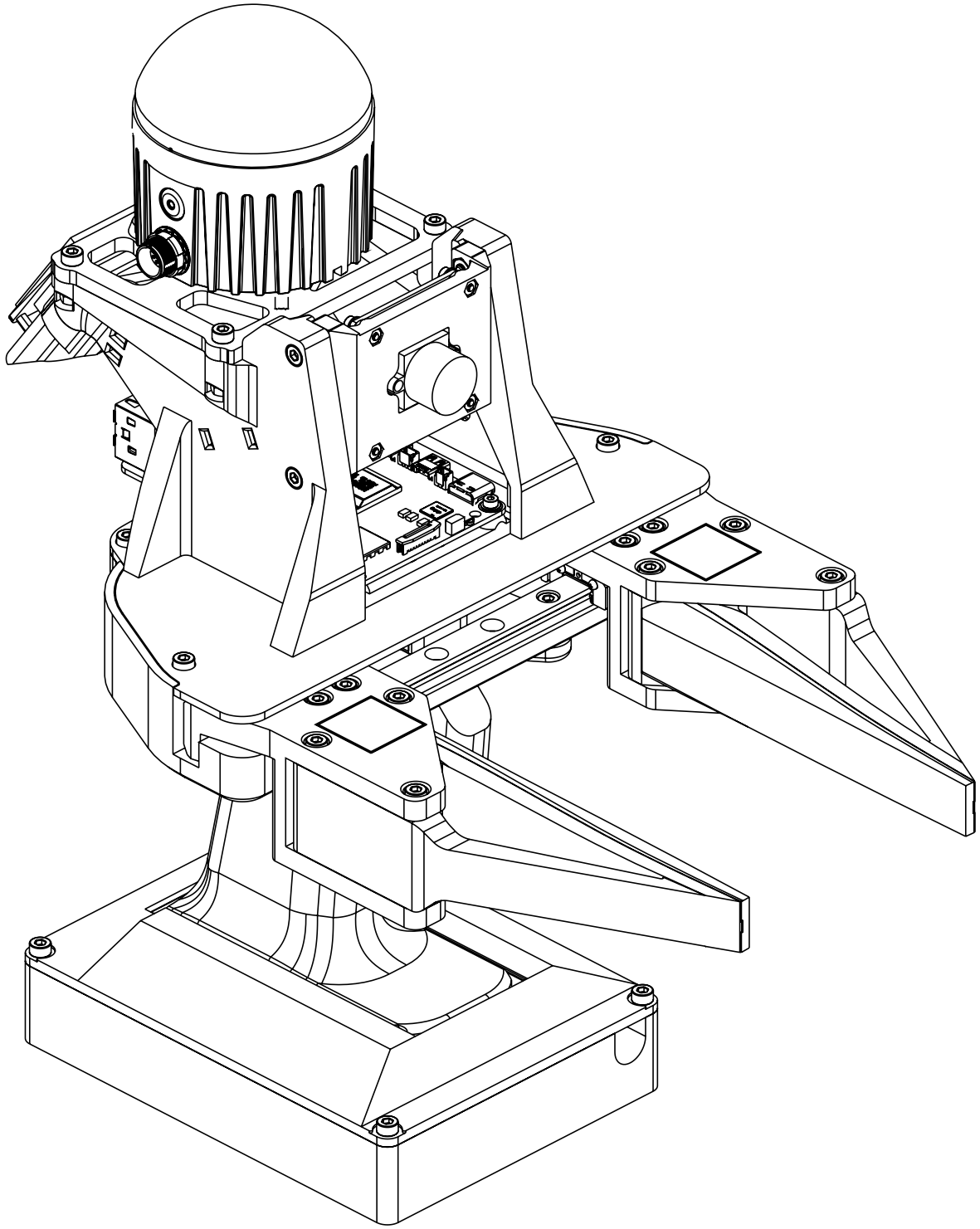
UMI-Live

Hardware Instructions



ASSEMBLY GUIDE

MECHANICAL / ELECTRONICS / WIRING

v9

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Notes

This guide assembles one UMI-Live gripper. Build two units for a bimanual setup.

Recommended tools: M2/M3 drivers, scissors, flush cutters, multimeter, and wire stripper.

3D prints: remove and clean off all support materials.

Carriages: select two carriages that slide the smoothest on the rail.

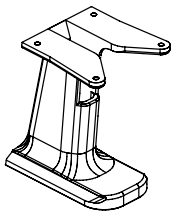
> **!** Keep carriages on a rail at all times to prevent the ball bearings from falling out.

Threadlocker: recommended for long-term or heavy use on metal-to-metal threads. Keep it off plastic, bearings, and sliding surfaces.

> **!** Keep the battery disconnected while assembling or changing wiring.

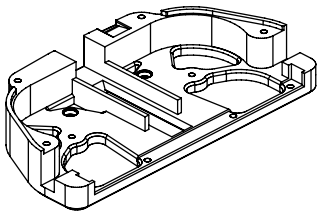
3D printed parts

Grip
grip.stl



A **1x**

Body bottom
body_bottom.stl



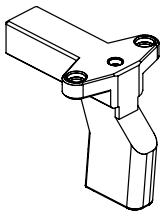
B **1x**

Gear axle
gear_axle.stl



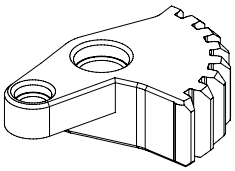
C **2x**

Trigger
trigger.stl



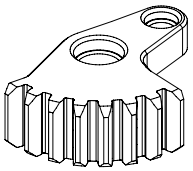
D **1x**

Left gear
gear_left.stl



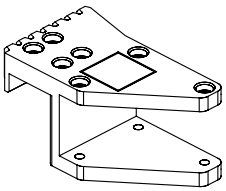
F **1x**

Right gear
gear_right.stl



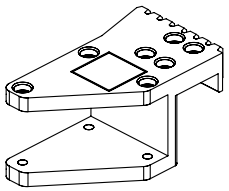
G **1x**

Left finger holder
finger_holder_left.stl



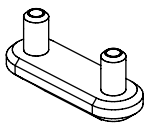
H-L **1x**

Right finger holder
finger_holder_right.stl



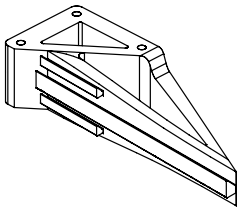
H-R **1x**

Linkage
linkage.stl



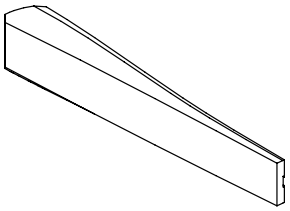
E **2x**

Finger backbone
finger_backbone.stl



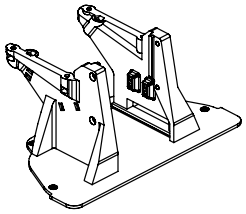
I **2x**

Finger pad
finger_pad.stl



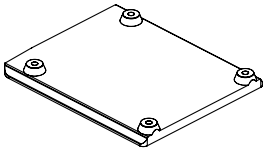
J **2x**

Body top
body_top.stl



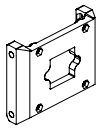
K **1x**

Pi slider
raspberry_slider.stl



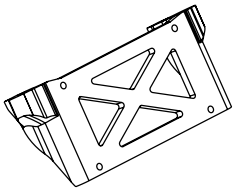
L **1x**

Camera mount
camera_mount.stl



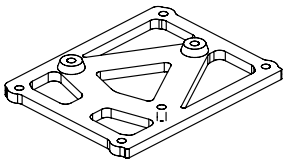
M **1x**

LCD mount
lcd_mount.stl



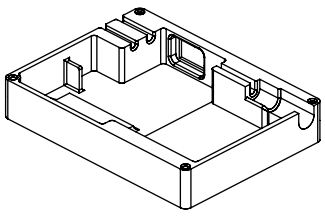
N **1x**

LiDAR mount
lidar_mount.stl



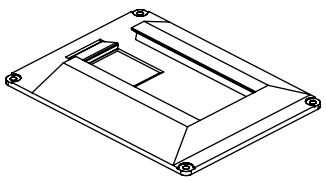
O **1x**

Battery case bottom
battery_pack_bottom.stl



P **1x**

Battery case top
battery_pack_top.stl

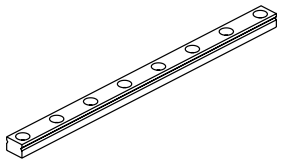


Q **1x**

Illustrations are not to scale.

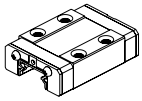
Mechanical parts

Linear rail
MGN9, 150 mm



R **1x**

Carriage block
MGN9C



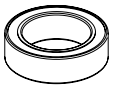
S **2x**

Small bearing
uxcell MR85-2RS



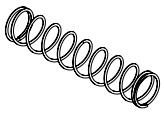
T **8x**

Large bearing
uxcell MR128-2RS



U **4x**

Compression spring
uxcell, 10 × 0.7 × 50 mm



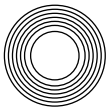
V **1x**

ArUco tags
DICT_4X4_50, 16 mm



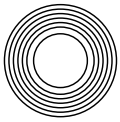
AM **2x**

Grip tape
3M TB641 gripping material, black



AN **Cut to fit**

Low-friction tape
UHMW polyethylene slick tape



AP **Cut to fit**

Consumables

Double-sided tape	**Zip ties**	**Blue threadlocker**
Cut to fit	**Optional**	**Optional**
Attaches the ArUco tags	Cable organization	Metal-to-metal threads only

Fasteners

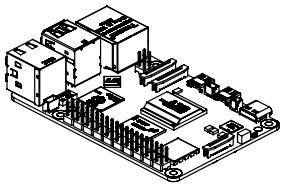
	H1 **M3 × 8 mm**	**14x**
	H2 **M3 × 12 mm**	**15x**
	H3 **M3 × 16 mm**	**2x**
	H4 **M3 × 20 mm**	**2x**
	H5 **M3 × 25 mm**	**10x**
	H6 **M3 × 30 mm**	**1x**
	H7 **M3 × 35 mm**	**6x**

	H8 **M3 nut**	**34x**
	H9 **M3 washer**	**2x**
	H10 **M2 × 6 mm**	**4x**
	H11 **M2 × 8 mm**	**8x**
	H12 **M2 nut**	**8x**
	H13 **M2 washer**	**12x**

Illustrations are not to scale.

Electronics and cables

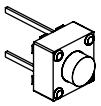
Raspberry Pi 5



W **1x**

Tactile button

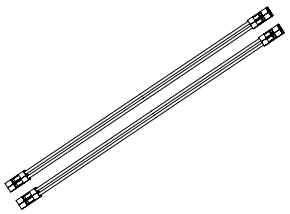
2-pin



X **2x**

Dupont jumper wires

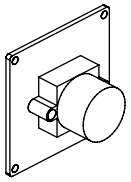
10 cm, female-to-female



AL **4x**

SVPRO USB camera

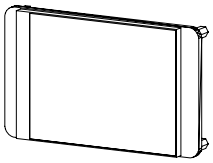
2MP global shutter



Y **1x**

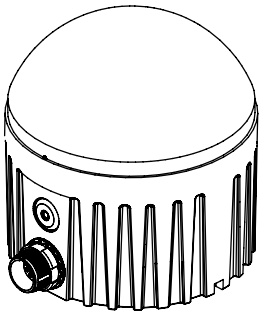
Waveshare LCD

2-inch SPI display, 240 × 320



Z **1x**

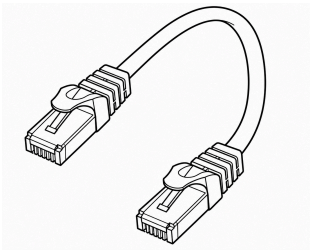
RoboSense Airy LiDAR



AA **1x**

Ethernet cable

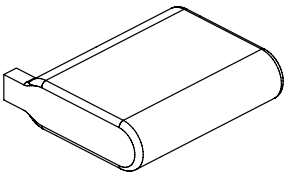
Short, 0.5 ft



AK **1x**

Battery

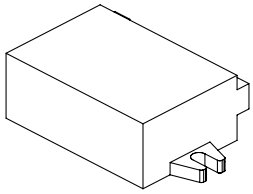
12 V 2600 mAh rechargeable Li-ion battery



AB **1x**

DC-DC converter

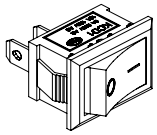
12/24 V to 5 V, 5 A, USB-C output



AC **1x**

Power switch

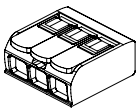
DaierTek mini rocker switch



AD **1x**

3-hole WAGO

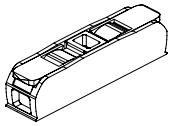
WAGO 221-413 lever nut



AF **2x**

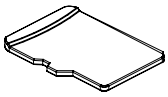
Inline WAGO

WAGO 221-2401 lever nut



AG **1x**

microSD card

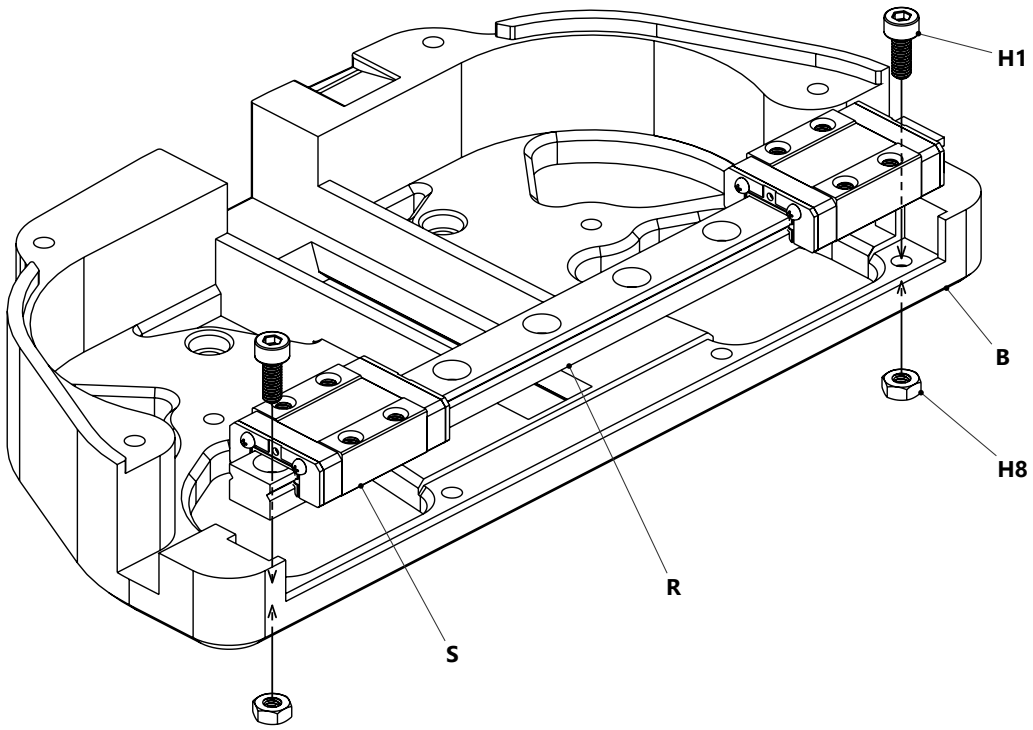


AH **1x**

Illustrations are not to scale.

STEP 1

Fit the rail and carriages



Select the two smoothest carriage blocks. Secure the rail to the body bottom using two M3 × 8 mm screws in the outer holes and two nuts underneath.

Keep both carriages on the rail to prevent the ball bearings from falling out.

PARTS ADDED

B **1x**
Body bottom

R **1x**
Linear rail

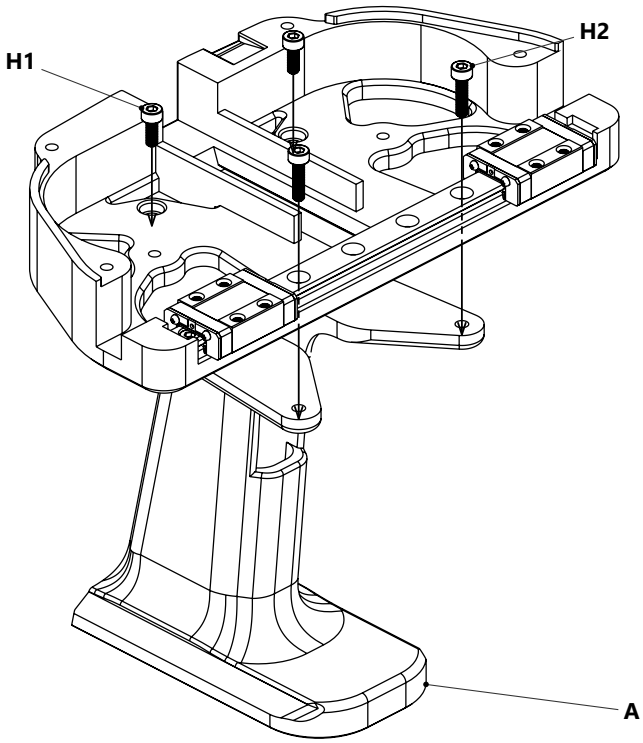
S **2x**
Carriage block
MGN9C

H1 **2x**
M3 × 8 mm

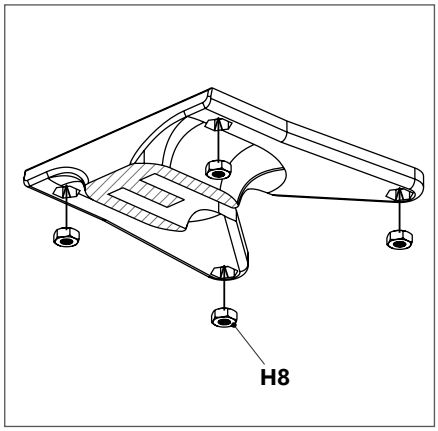
H8 **2x**
M3 nut

STEP 2

Attach the grip



UNDERSIDE VIEW



Align the grip beneath the body bottom. Fit two M3 × 12 mm screws through the inner rail holes, body bottom, and grip, plus two M3 × 8 mm screws through the other grip mounting holes. Secure all four screws with nuts underneath and tighten evenly.

PARTS ADDED

A **1x**
Grip

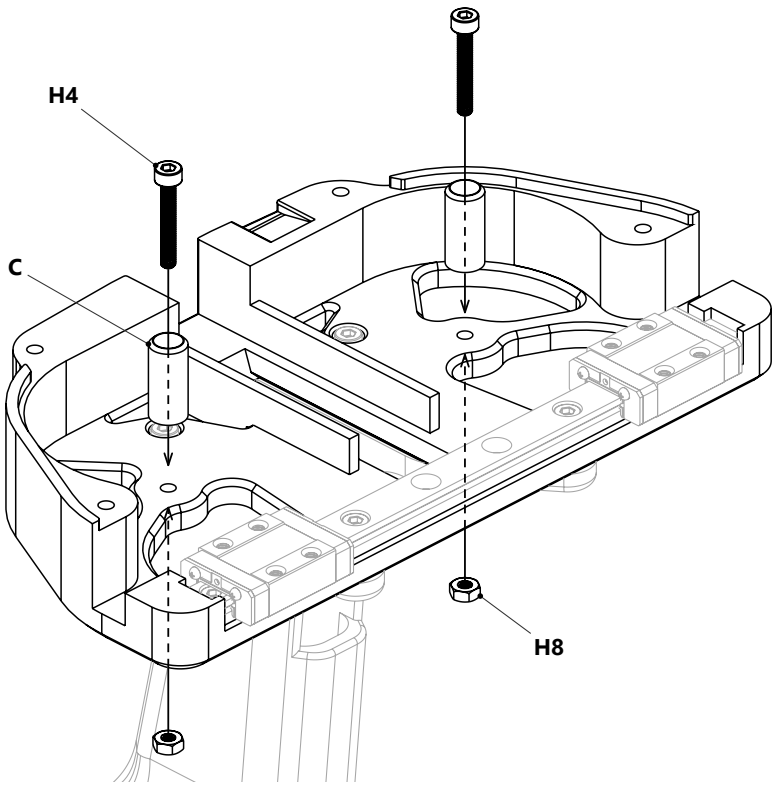
H1 **2x**
M3 × 8 mm

H2 **2x**
M3 × 12 mm

H8 **4x**
M3 nut

STEP 3

Install the gear axles



Secure each gear axle with one M3 × 20 mm screw and nut. Ensure that both axles are seated vertically.

PARTS ADDED

C **2x**

Gear axle

H4 **2x**

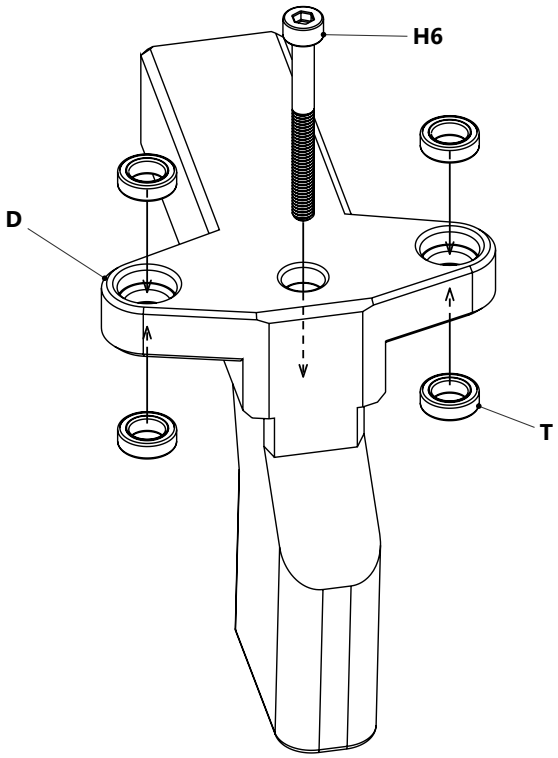
M3 × 20 mm

H8 **2x**

M3 nut

STEP 4

Prepare the trigger



Seat one small bearing on each face of both trigger pivots (four total). Screw the M3 × 30 mm screw into the central hole, threading the plastic.

Do not force the bearings into their recesses. If the fit is too tight, carefully file to adjust the trigger's bearing holes until the bearings seat without excessive force.

PARTS ADDED

D **1x**

Trigger

T **4x**

Small bearing

MR85-2RS

H6 **1x**

M3 × 30 mm

STEP 5

Apply low-friction tape

PARTS ADDED

AP **Cut to fit**

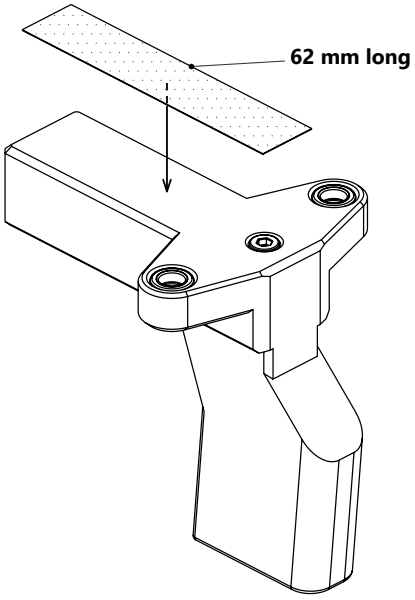
Low-friction tape

ALREADY ASSEMBLED

Prepared trigger

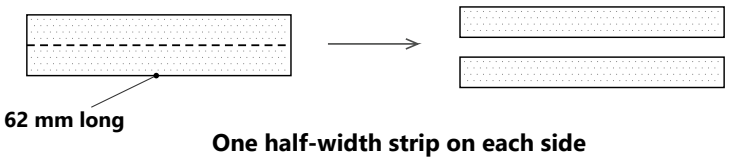
From Step 4

1. Top strip

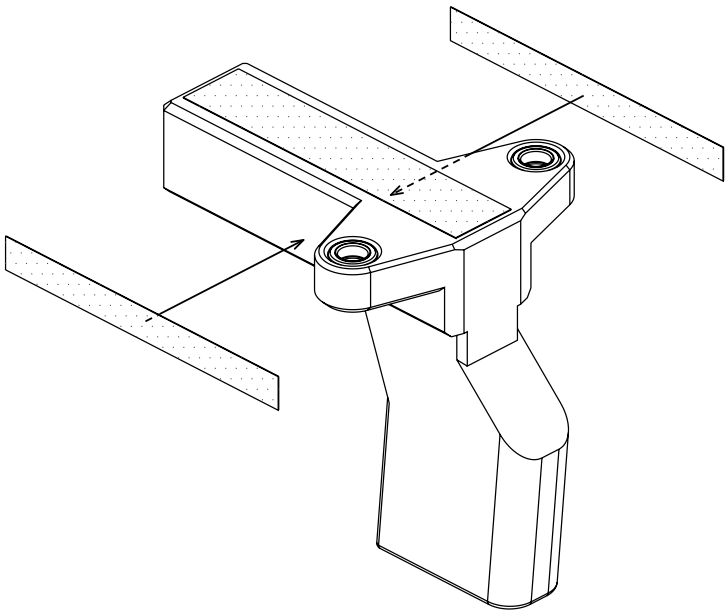


Cut a 62 mm long strip of low-friction tape. Apply it along the top of the trigger, covering the recessed screw.

2. Side strips



One half-width strip on each side



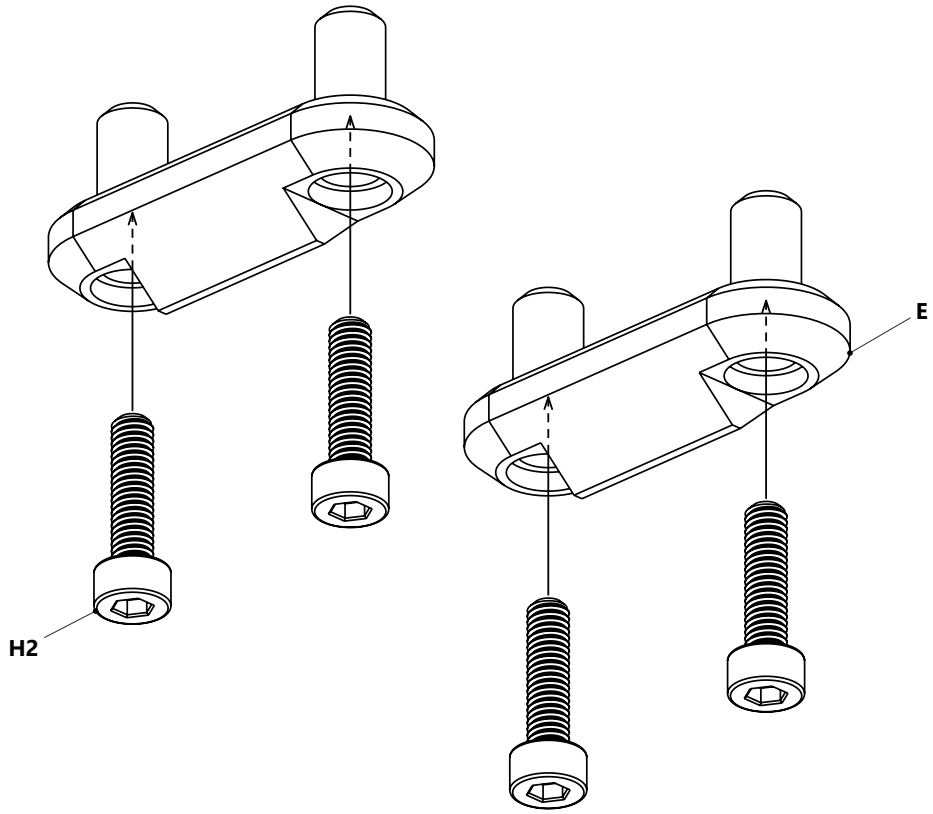
Cut another 62 mm long strip of low-friction tape, and cut it lengthwise. Apply one half-width strip to each side of the trigger.

Press firmly over all applied tape to secure it.

Do not stack or overlap tape layers. Apply tape only to flat surfaces; do not bend around corners.

STEP 6

Prepare the linkages



Screw two M3 × 12 mm screws into each linkage from its underside, threading the plastic.

PARTS ADDED

E **2x**

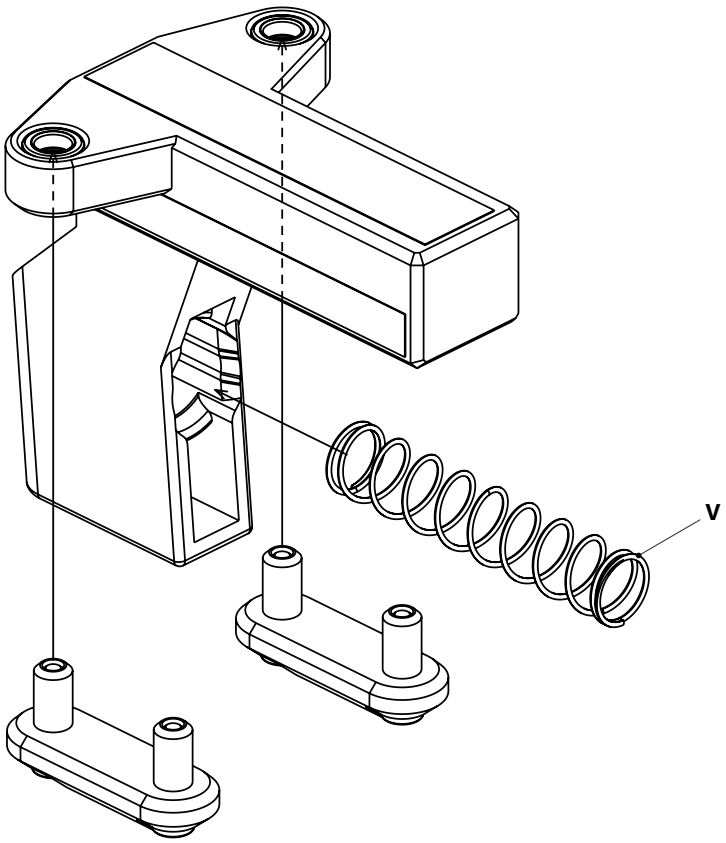
Linkage

H2 **4x**

M3 × 12 mm

STEP 7

Join the trigger and linkages



Bring both prepared linkages up from below into the trigger bearings. Slide the compression spring into the cylindrical slot in the trigger, along the axis shown.

PARTS ADDED

V **1x**

Compression spring

ALREADY ASSEMBLED

Prepared trigger

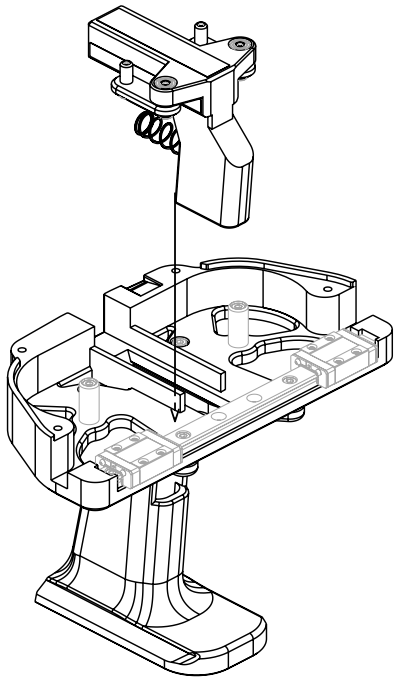
From Step 5

Prepared linkages

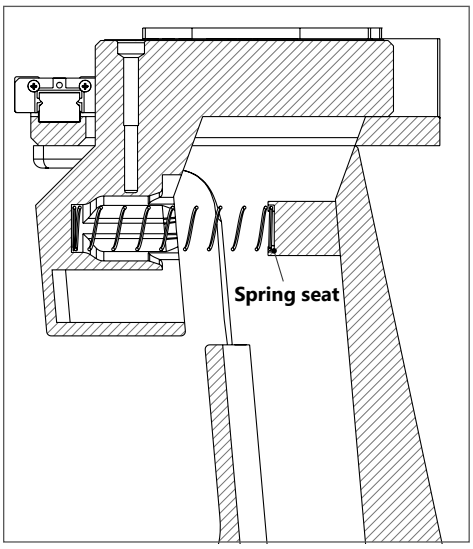
From Step 6

STEP 8 Install the trigger

LOWER INTO BODY



CROSS-SECTIONAL VIEW



Lower the trigger into its slot in the body assembly. Position the spring into its seat in the grip. Ensure both linkages remain fully seated in their bearings.

Tweezers may be needed to position the spring correctly in its seat.

ALREADY ASSEMBLED

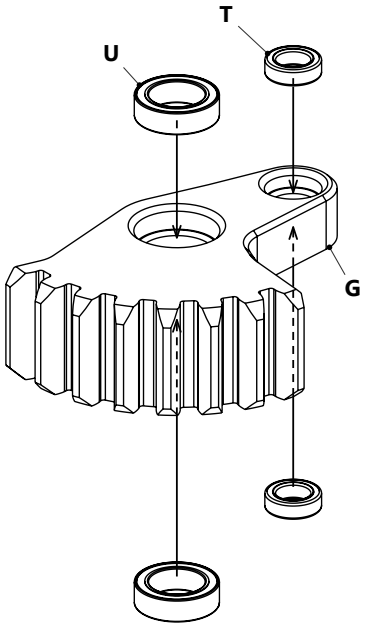
Body assembly

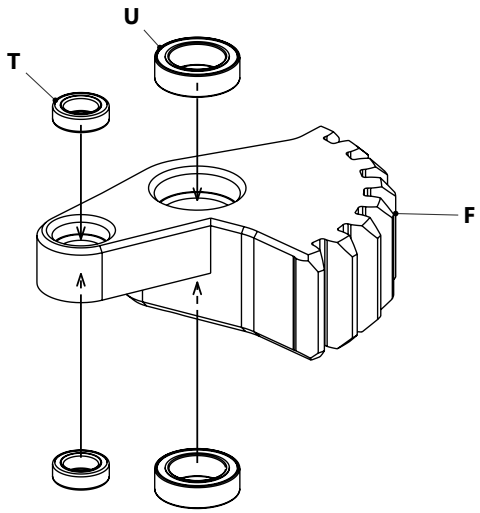
From Step 3

Trigger + linkages

From Step 7

STEP 9 Prepare the gears





Fit two large MR128-2RS bearings into the main pivot and two small MR85-2RS bearings into the linkage pivot of each gear. Ensure that bearings are seated squarely in their recesses.

Do not force the bearings into their recesses. If the fit is too tight, carefully file to adjust the gear holes until the bearings seat without excessive force.

PARTS ADDED

F **1x**

Left gear

G **1x**

Right gear

U **4x**

Large bearing

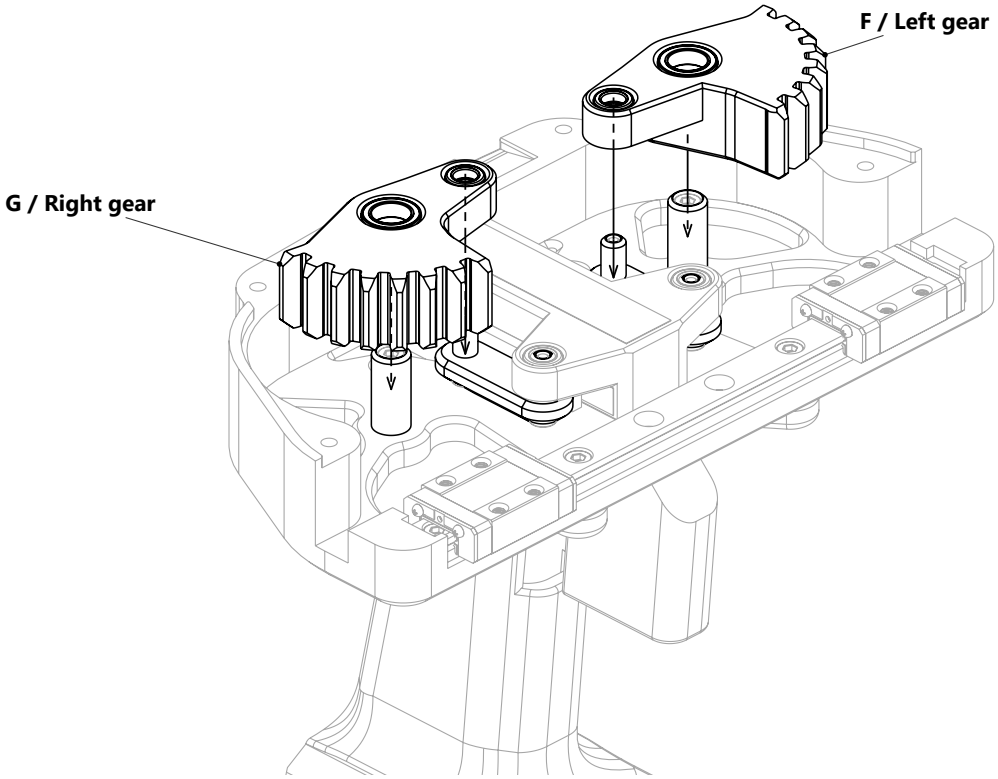
MR128-2RS

T **4x**

Small bearing

MR85-2RS

STEP 10 Install the gears



Place the prepared gears over the two printed axles. Ensure that all bearings are fitted into their axles. Move the trigger gently to check that both sides travel together smoothly.

ALREADY ASSEMBLED

Body assembly

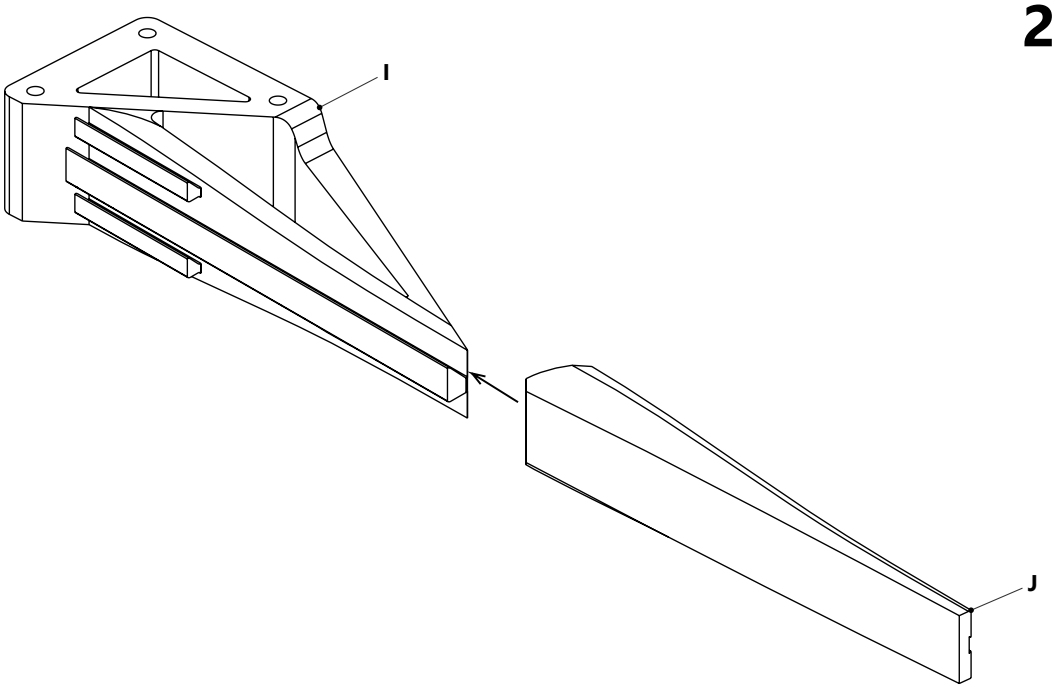
From Step 8

Prepared gears

From Step 9

STEP 11 Fit the finger pads

2x



Slide the TPU pad onto its backbone from the finger tip. Keep it aligned with the backbone as you push/pull it along the full length. Build another set for a total of two fingers.

The pad slides on from the front; do not push it on sideways.

PARTS ADDED

I **2x**

Finger backbone

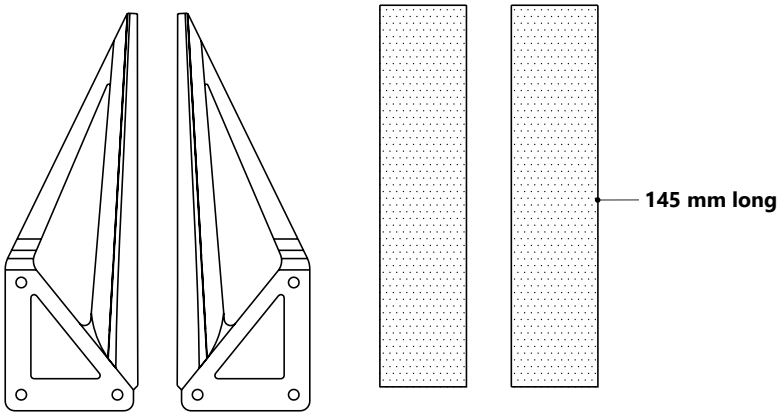
J **2x**

Finger pad

STEP 12

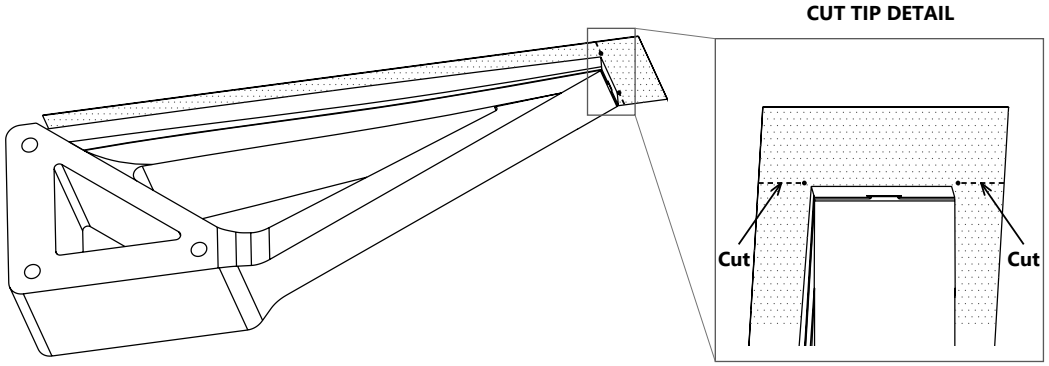
Apply the grip tape

1. Ready for tape



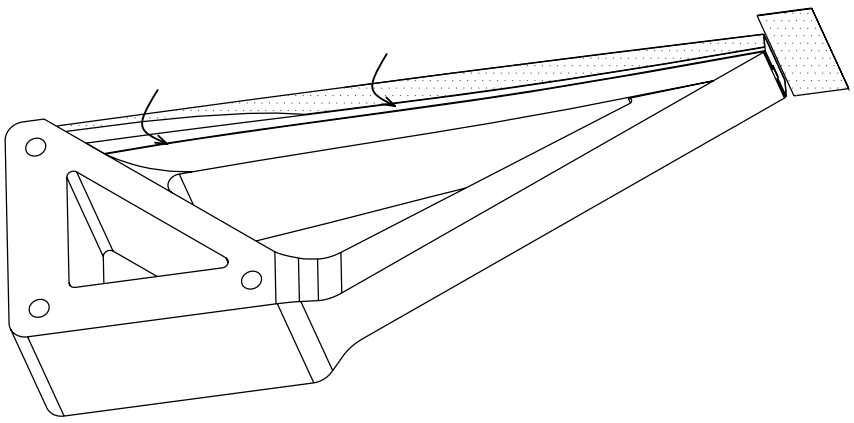
Start with both padded fingers from Step 11. Cut two strips of grip tape, each around 145 mm long.

2. Apply the tape and cut the tip



Center the tape on the TPU contact face, leaving overhangs along both long edges and the tip. Cut inward from each tape edge to the tip corner (detail), leaving the center uncut.

3. Fold the long edges



Fold both long edges around the TPU pad and press them down. Keep the tip tab free for the next substeps.

PARTS ADDED

AN **Cut to fit**

Grip tape

ALREADY ASSEMBLED

Finger pair

From Step 11

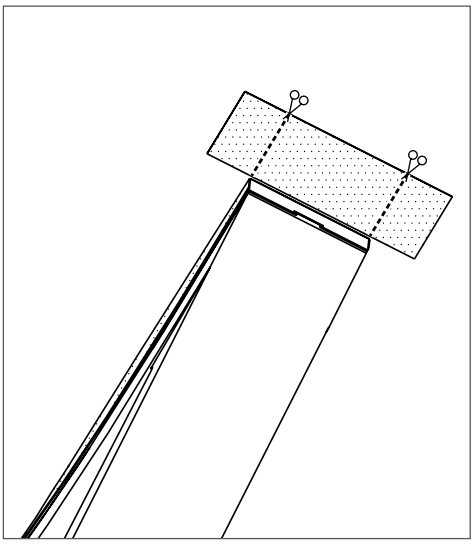
STEP 12

Finish the grip tape (continued)

CONTINUE WITH

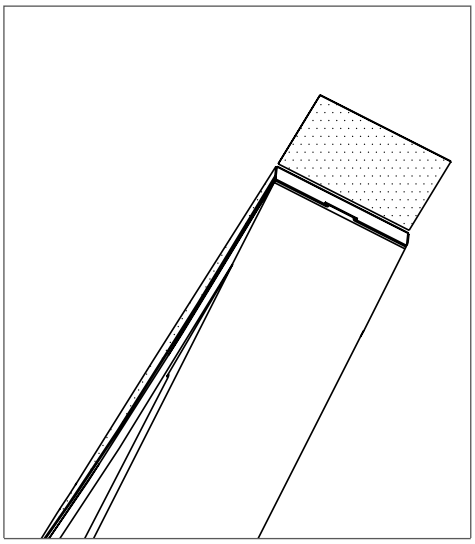
Same fingers and grip tape from the previous page.

4. Trim the tip sides



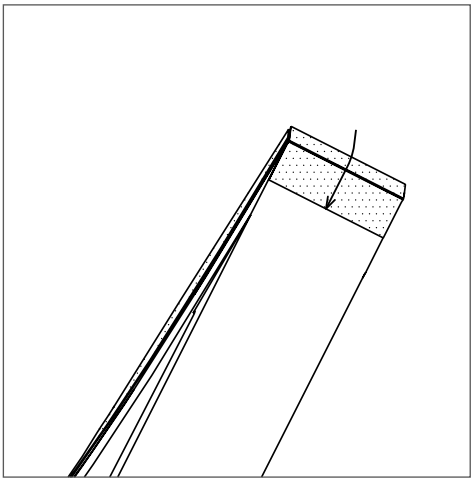
Trim the tab sides along the dashed lines so that each cut is aligned with an edge of the finger.

5. Tab trimmed to width



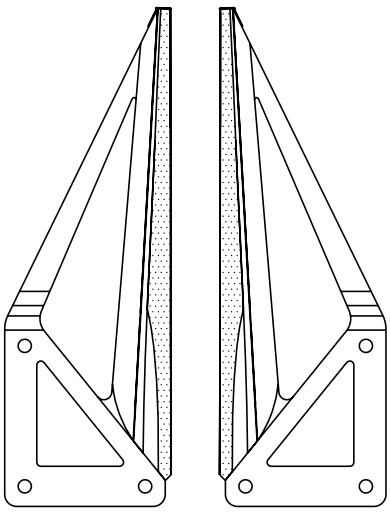
Check that the tab now matches the width of the finger. Ensure this flap is long enough to fold over the tip.

6. Fold over the tip



Fold the flap over the tip and press it flat against the back of the finger.

7. Finished pair



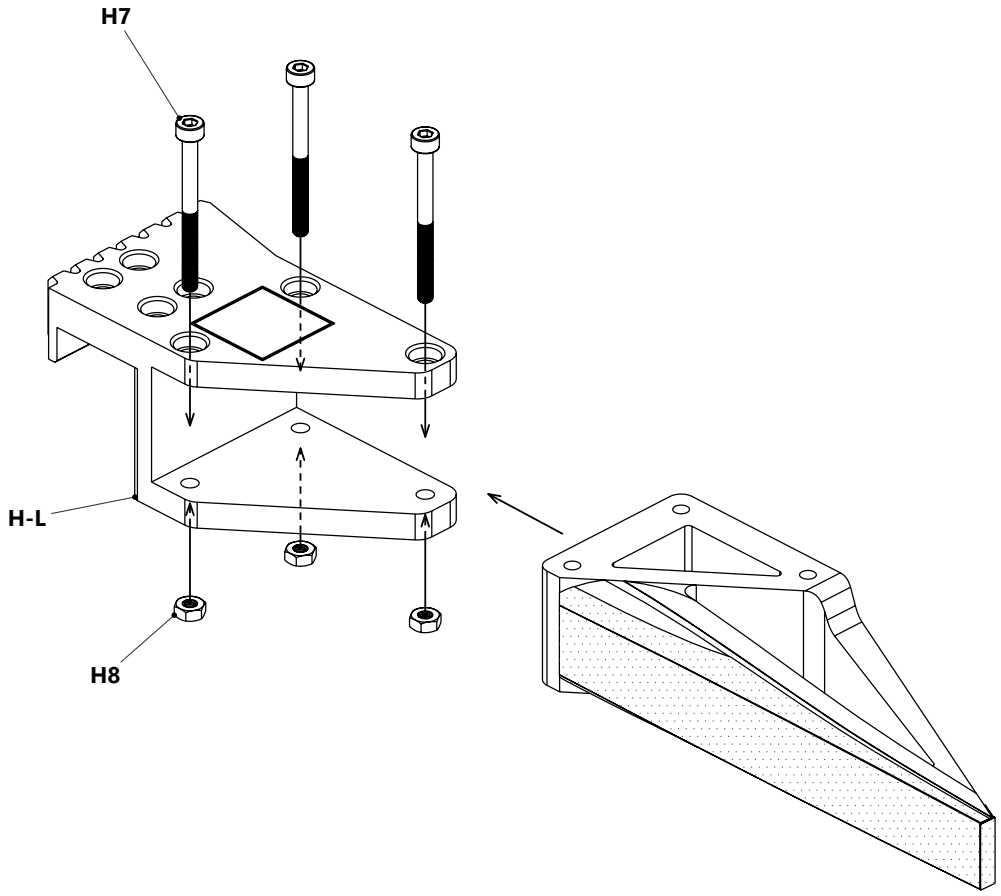
Repeat for the other finger.

Press firmly over all applied tape to secure it.

STEP 13

Mount the fingers

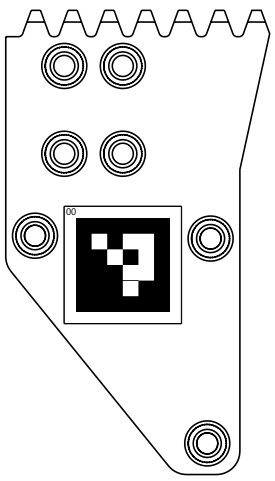
2x



Slide one completed finger into the left holder (H-L), as shown. Align the three holes and fit three M3 × 35 mm screws and three nuts. Repeat with the right holder (H-R) and the other finger.

Fit the ArUco tags

Print the supplied tag sheet at 100% (Actual size), with Fit and Shrink disabled. Check the 50 mm scale before cutting.



Dictionary: DICT_4X4_50

Black marker: 16 × 16 mm

Cut size: 20 × 20 mm; keep the 2 mm white margin.

Attach one tag inside each holder's recessed border using double-sided tape. Ensure the tag is flat and secured.

Use two distinct IDs; assign left and right in software.

PARTS ADDED

H-L **1x**
Left finger holder

H-R **1x**
Right finger holder

H7 **6x**
M3 × 35 mm

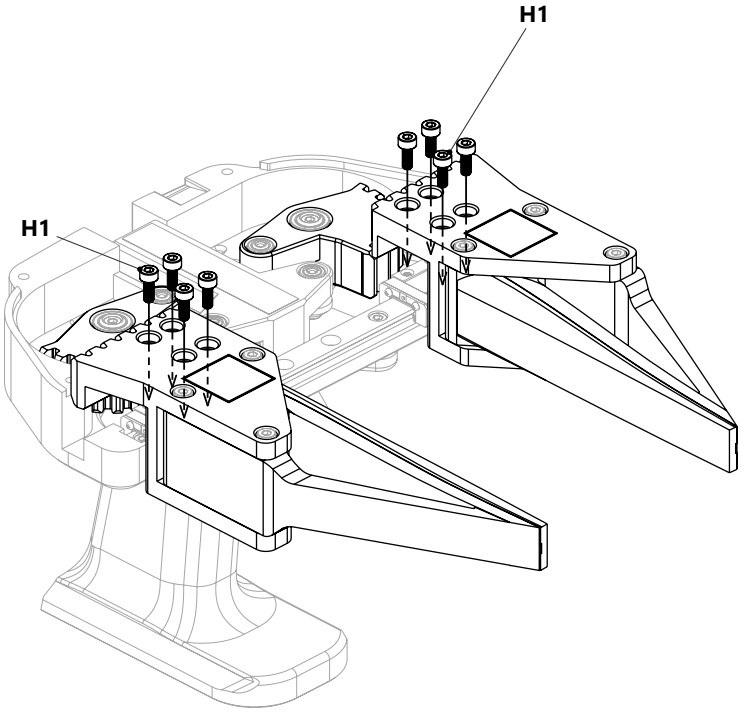
H8 **6x**
M3 nut

AM **2x**
ArUco tags

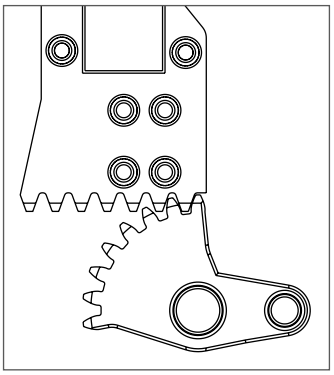
ALREADY ASSEMBLED

Finger pair
From Step 12

STEP 14 Install the fingers



TOP VIEW OF LEFT SIDE



Mesh the left holder (H-L) with the left gear (F), and the right holder (H-R) with the right gear (G). Align each holder over its carriage and secure it with four M3 × 8 mm screws. Check that both fingers move symmetrically.

PARTS ADDED

H1 **8x**

M3 × 8 mm

ALREADY ASSEMBLED

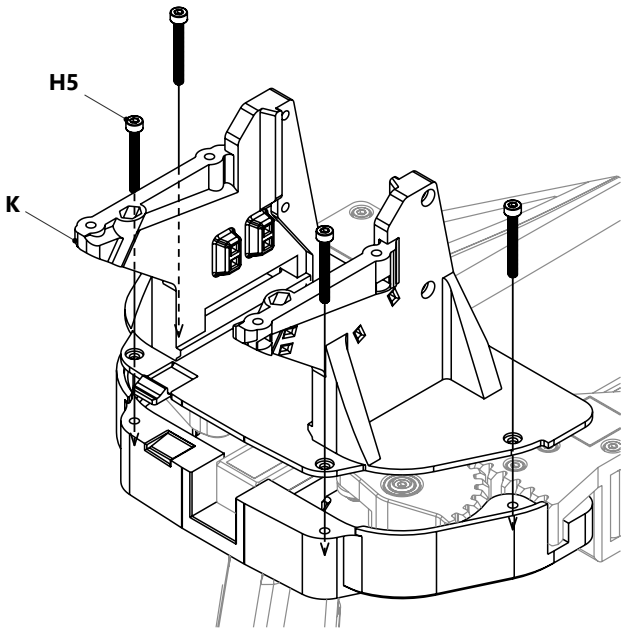
Body assembly

From Step 10

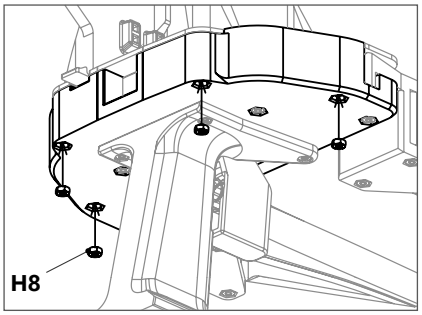
Finger assemblies

From Step 13

STEP 15 Close the body



UNDERSIDE VIEW



Lower the body top over the completed mechanism. Fit four M3 × 25 mm screws and nuts. Tighten gradually, checking the trigger and finger movement to prevent binding.

PARTS ADDED

K **1x**

Body top

H5 **4x**

M3 × 25 mm

H8 **4x**

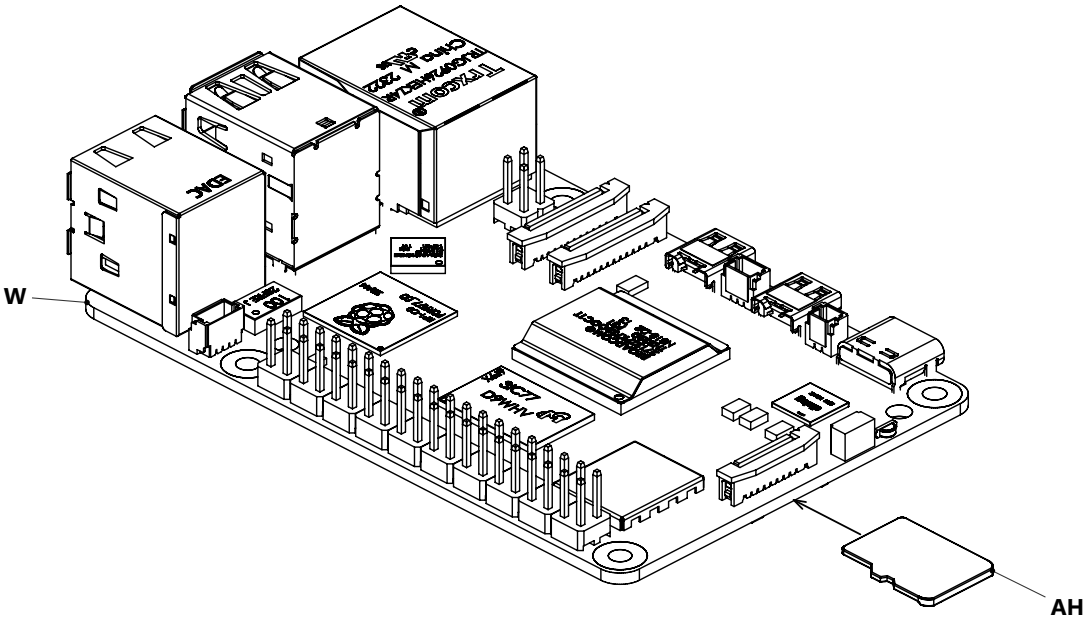
M3 nut

ALREADY ASSEMBLED

Body assembly

From Step 14

STEP 16 Prepare the Pi



Prepare the microSD card using the UMI-Live software setup guide before first use. Software setup can be completed after assembly.

Insert the microSD card into the underside slot, with the gold contacts facing up toward the board.

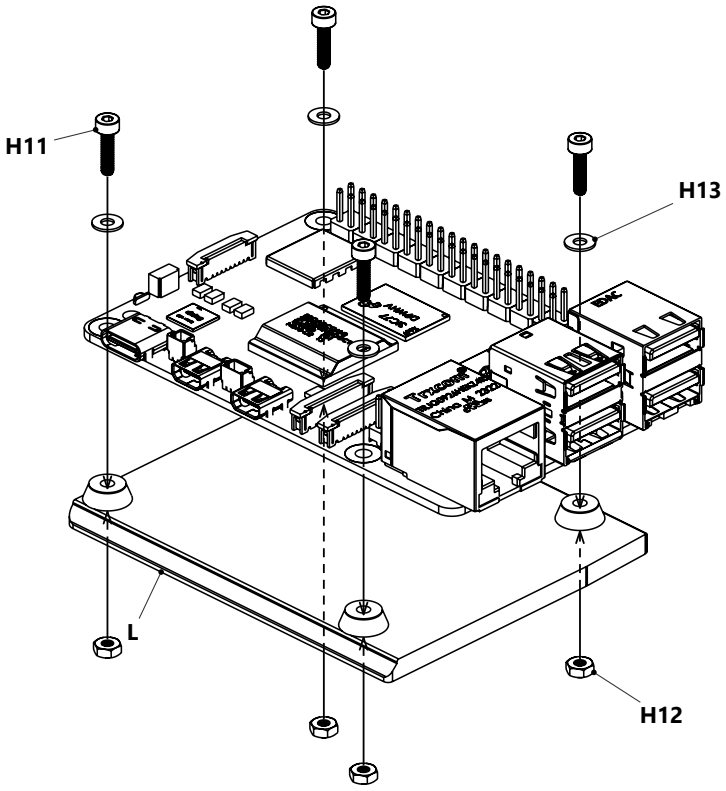
Slide it in gently; do not force it.

PARTS ADDED

W **1x**
Raspberry Pi 5

AH **1x**
microSD card

STEP 17 Mount the Pi



Place the Raspberry Pi on its slider and align the four mounting holes. Fit four M2 × 8 mm screws, washers, and nuts. Tighten just enough to secure the Raspberry Pi. Do not overtighten.

Note the orientation of the slider and Pi.

PARTS ADDED

L **1x**
Pi slider

H11 **4x**
M2 × 8 mm

H12 **4x**
M2 nut

H13 **4x**
M2 washer

ALREADY ASSEMBLED

Prepared Pi
From Step 16

STEP 18

Connect the Pi

1. Plug the included cable into the LCD. Trace each wire from its label on the LCD to its loose end.
2. Connect all eight LCD wires to the Pi using the diagram below, then unplug the LCD end.
3. Connect four female-to-female Dupont leads to the Pi for BTN 1 and BTN 2.

Note that GPIO numbers are different from physical pin numbers.

Keep power disconnected.

PARTS ADDED

AL **4x**

Dupont jumper wires

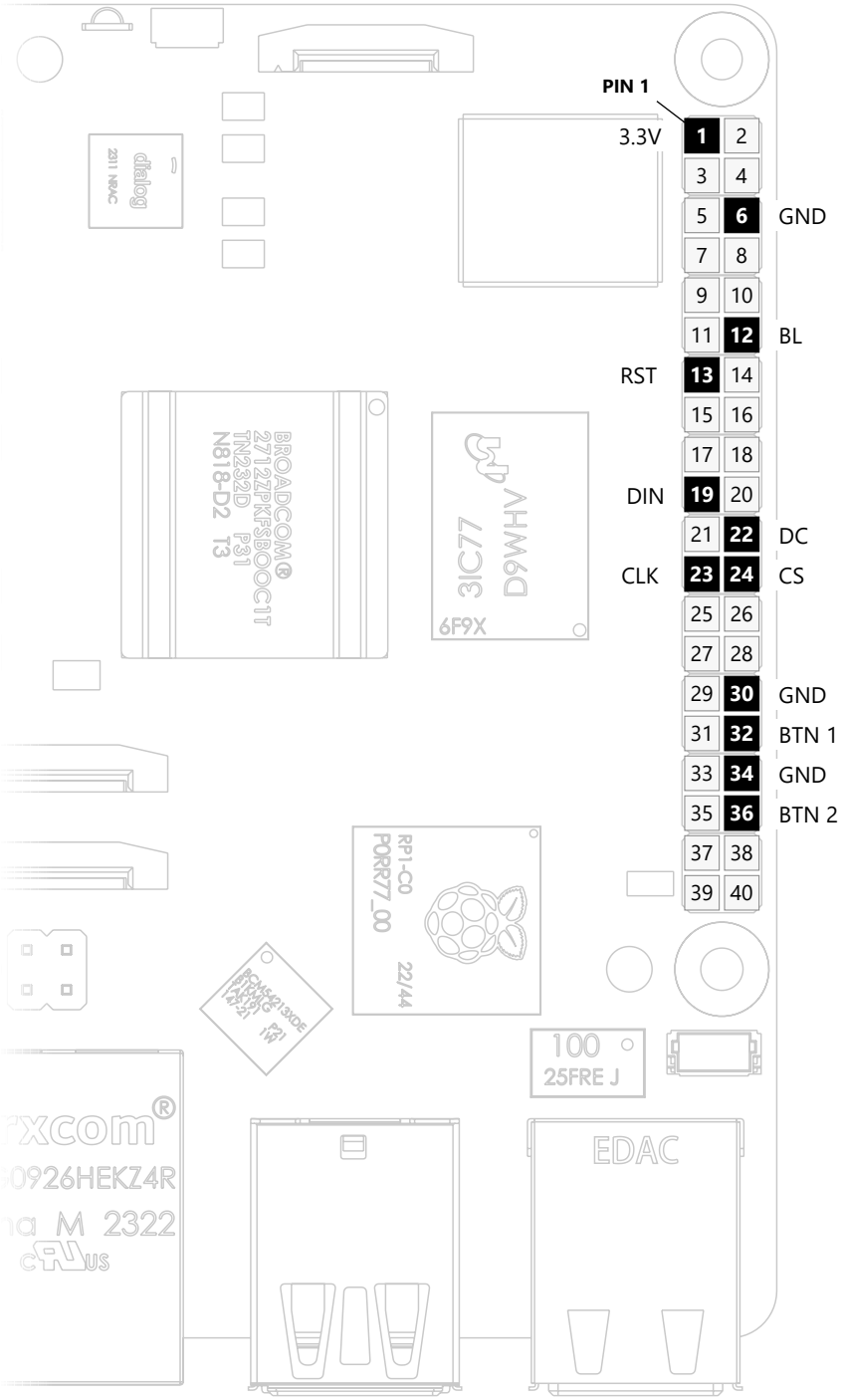
Female-to-female

1x

LCD cable

Included with LCD

PIN HEADER



CONNECTIONS

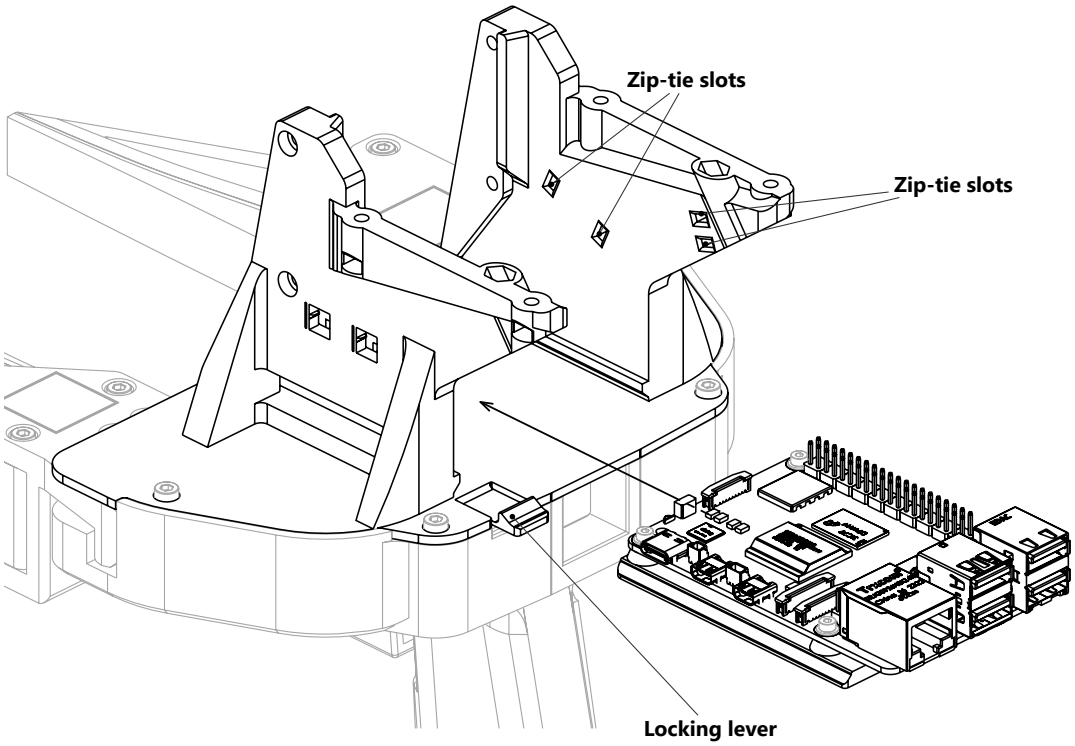
LCD	PI SIGNAL	PIN
VCC	3.3V	1
GND	Ground	6
DIN	GPIO10	19
CLK	GPIO11	23
CS	GPIO8	24
DC	GPIO25	22
RST	GPIO27	13
BL	GPIO18	12

BTN 1	PI SIGNAL	PIN
Signal	GPIO12	32
GND	Ground	30

BTN 2	PI SIGNAL	PIN
Signal	GPIO16	36
GND	Ground	34

Wires are generally not shown in the CAD illustrations.

STEP 19 Install the Pi



Insert the Pi and slider forward into the body top guides until the lever clicks and locks. Thread optional zip ties through the marked slots and around the LCD cables; leave them loose until wiring is complete.

PARTS ADDED

2 optional

Zip ties

ALREADY ASSEMBLED

Body assembly

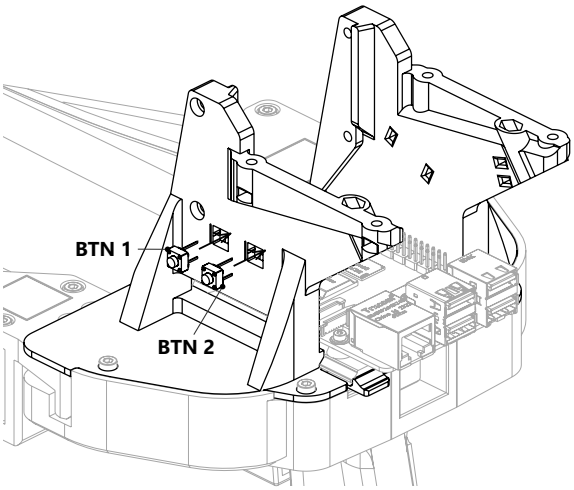
From Step 15

Pi + slider

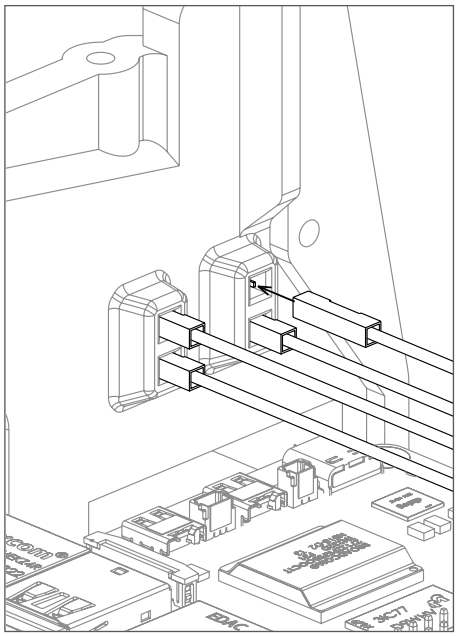
From Step 17

STEP 20 Install the buttons

FIT THE BUTTONS



CONNECT THE WIRES



Fully press the buttons in the side openings; the tabs are meant to bend to tightly fit the buttons. Connect the button nearest to the fingers to GPIO12 and GND (physical pins 32 and 30). Connect the button nearest to the LCD to GPIO16 and GND (physical pins 36 and 34). Refer to the diagram in Step 18 as needed.

PARTS ADDED

X **2x**

Tactile button

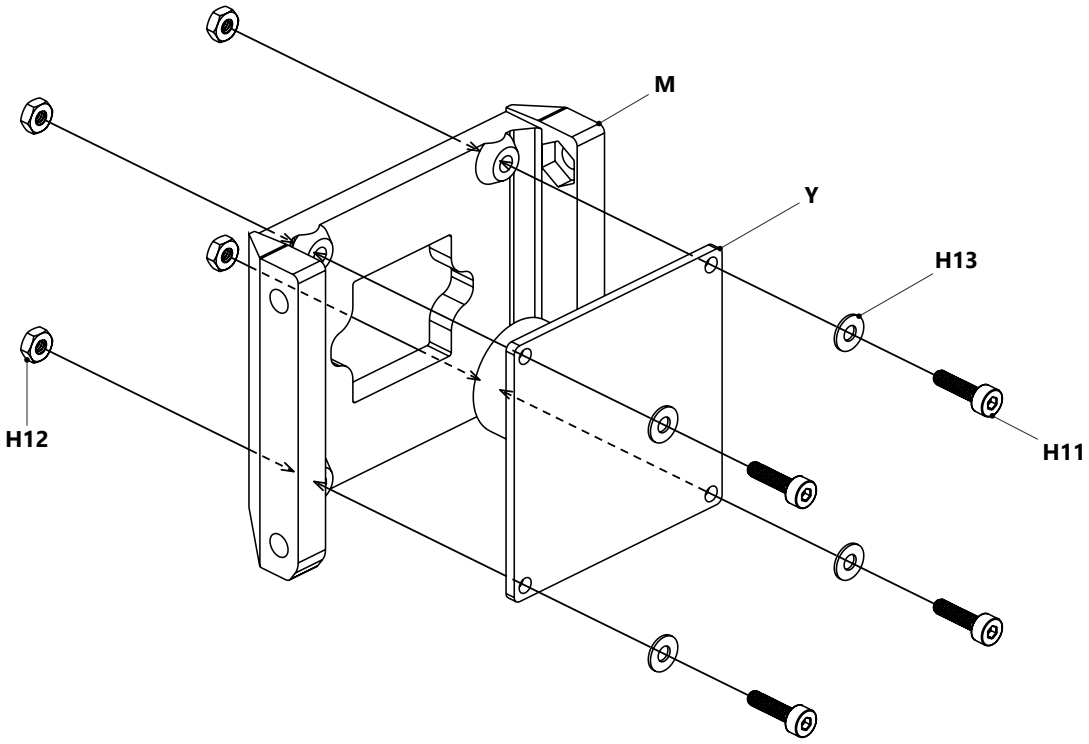
ALREADY ASSEMBLED

Body assembly

From Step 19

STEP 21

Mount the camera



Bring the camera board forward from behind the mount, with the lens facing through the opening. Fit four M2 × 8 mm screws, washers, and nuts. Connect the 4-pin end of the USB camera cable to the camera board.

PARTS ADDED

M **1x**
Camera mount

Y **1x**
SVPRO USB camera

1x
USB camera cable
Included with camera

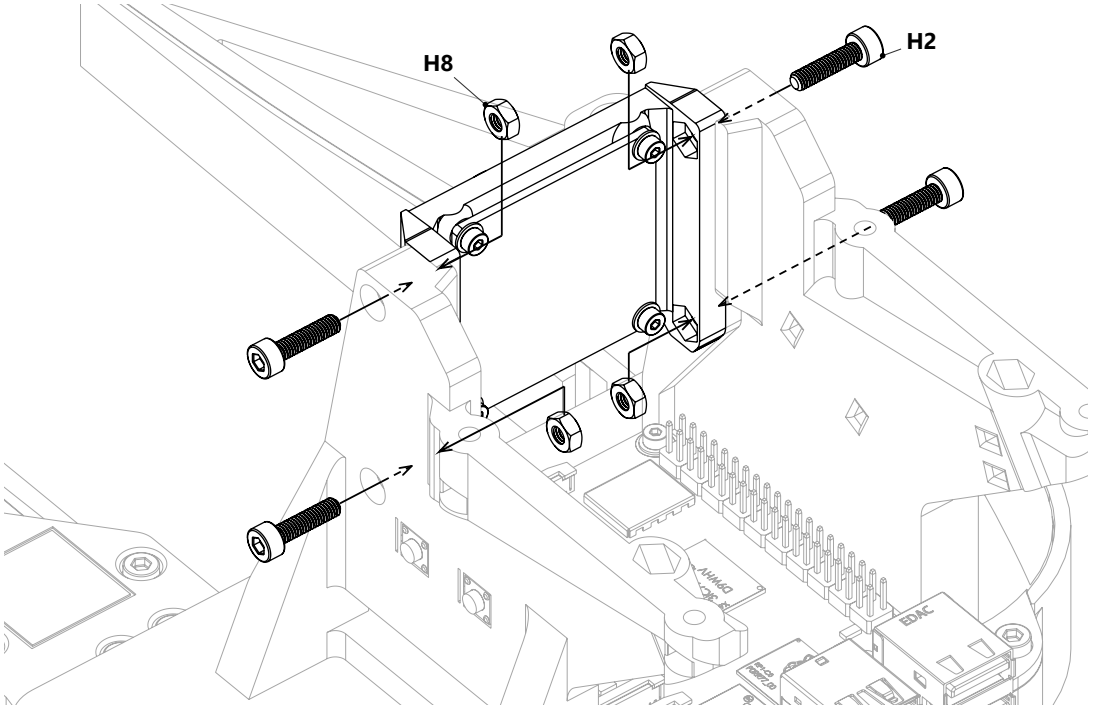
H11 **4x**
M2 × 8 mm

H12 **4x**
M2 nut

H13 **4x**
M2 washer

STEP 22

Install the camera



Secure the camera mount to the front of the body top with four M3 × 12 mm screws and nuts. Plug the USB camera cable into a blue USB-A 3.0 port on the Raspberry Pi.

PARTS ADDED

H2 **4x**
M3 × 12 mm

H8 **4x**
M3 nut

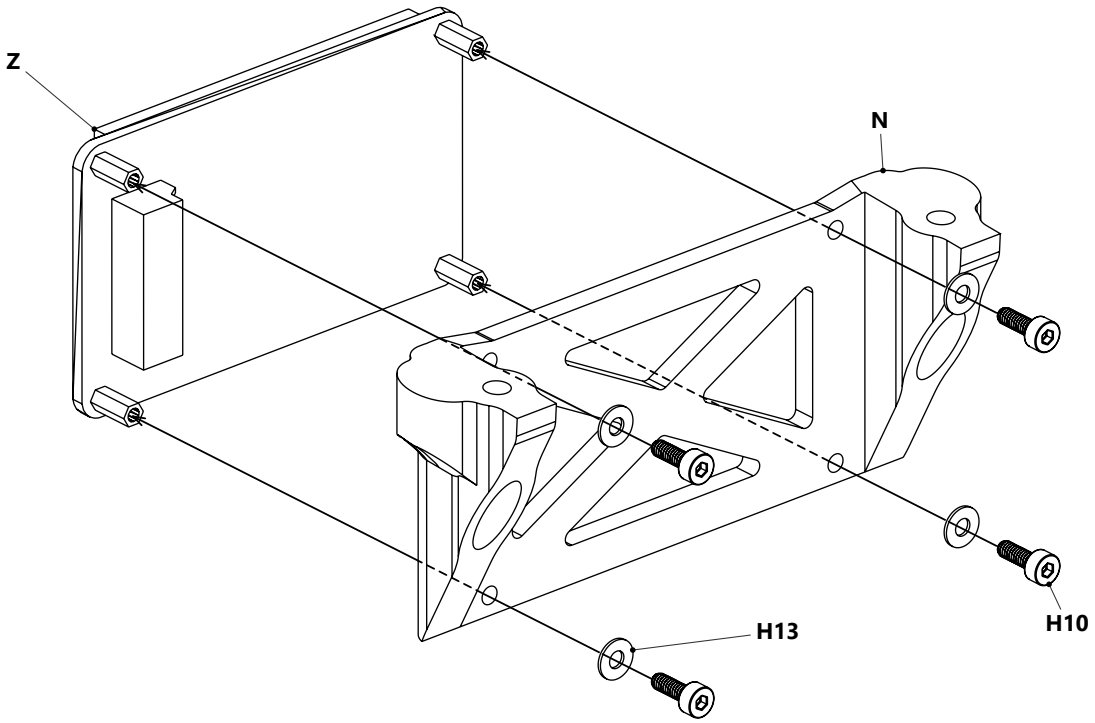
ALREADY ASSEMBLED

Body assembly
From Step 20

Camera assembly
From Step 21

STEP 23

Mount the LCD



Align the LCD onto the printed mount with the display facing outward. Fit four M2 × 6 mm screws and washers.

PARTS ADDED

N **1x**

LCD mount

Z **1x**

Waveshare LCD

H10 **4x**

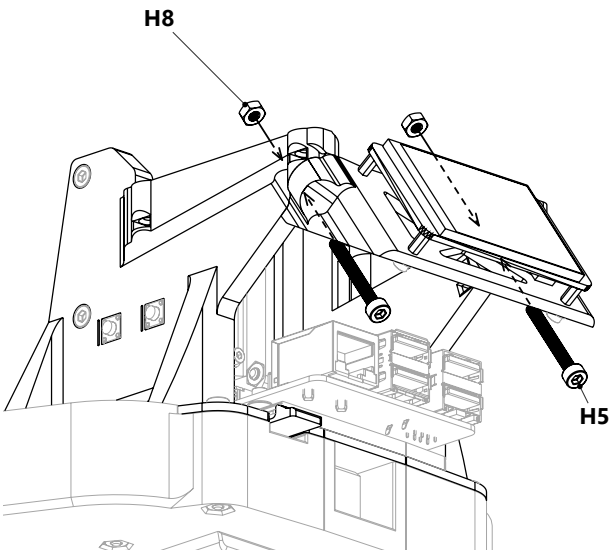
M2 × 6 mm

H13 **4x**

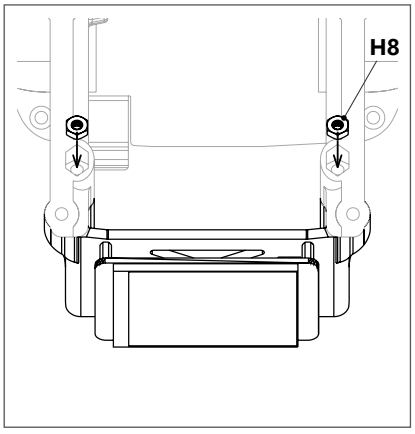
M2 washer

STEP 24

Install the LCD



TOP VIEW



Mount the LCD assembly at the rear of the body using two M3 × 25 mm screws and nuts. Connect the LCD cable from the Pi to the LCD display. It is recommended to recheck each cable before applying power. Tighten and cut the optional zip ties as needed.

Ensure that the LCD cable is fully pushed into the LCD display.

PARTS ADDED

H5 **2x**

M3 × 25 mm

H8 **2x**

M3 nut

ALREADY ASSEMBLED

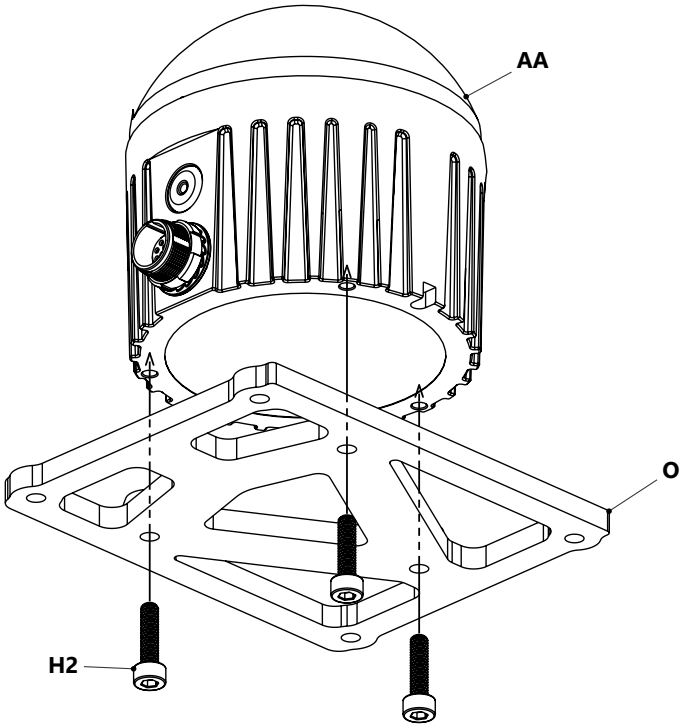
Body assembly

From Step 22

LCD assembly

From Step 23

STEP 25 Mount the LiDAR



Align the Airy LiDAR base with the LiDAR mount. Note the orientation of the cable connector on the LiDAR. Screw three M3 × 12 mm screws through the plate into the LiDAR.

Avoid touching the LiDAR optical dome. Keep the dome and ventilation clear.

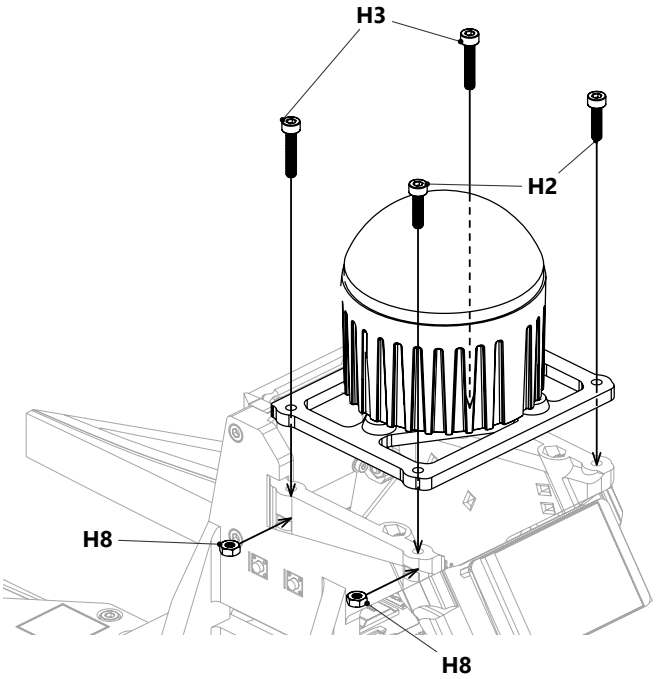
PARTS ADDED

O **1x**
LiDAR mount

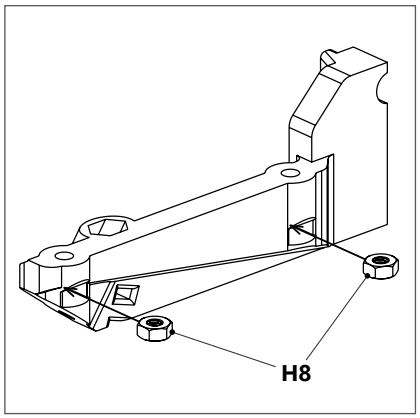
AA **1x**
RoboSense Airy LiDAR

H2 **3x**
M3 × 12 mm

STEP 26 Install the LiDAR



OPPOSITE SIDE



Slide four M3 nuts into the four slots, two inward from each outer side. Place the LiDAR mount onto the body with the cable connector on the LiDAR facing right. Fit two M3 × 16 mm screws at the front holes (camera side) and two M3 × 12 mm screws at the rear (LCD side).

PARTS ADDED

H2 **2x**
M3 × 12 mm

H3 **2x**
M3 × 16 mm

H8 **4x**
M3 nut

ALREADY ASSEMBLED

Body assembly
From Step 24

LiDAR assembly
From Step 25

STEP 27 Prepare the battery pack wires

Shortening is optional and makes the battery pack more organized. If shortening, cut to the lengths in the table below.

Strip 11 mm from the end of each wire lead, then twist the bare strands. Optionally solder the tip to prevent fraying.

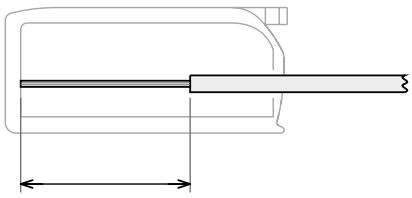
In addition to the cables in the table below, strip the leads of the male barrel jack that is included with the battery and charger. This jack will power the LiDAR.

> **!** Do not short-circuit the battery! Keep the insulator tubes on the bare wire leads when not in use.

COMPONENT	LEAD	CUT LENGTH
Battery	Red	29 mm
	Black	85 mm
Switch	Lead 1	28 mm
	Lead 2	49 mm
Converter	Red	65 mm
	Black	63 mm

Total cut length includes the 11 mm stripped end.

WAGO SIDE VIEW



11 mm stripped

PARTS ADDED

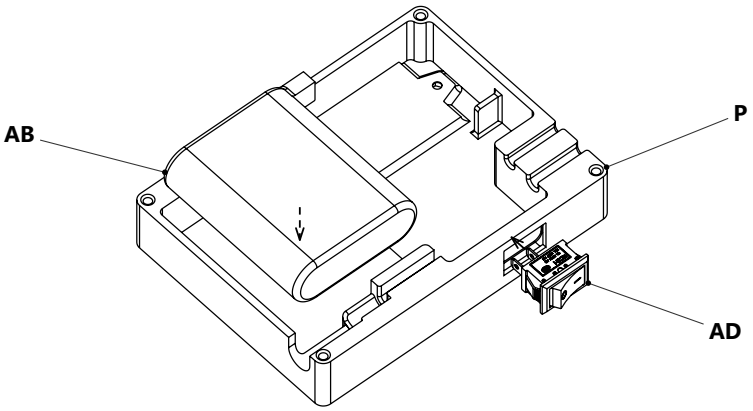
Male barrel jack **1x**
With battery and charger

Battery cables **2x**
On the battery (AB)

Converter cables **2x**
On the converter (AC)

Switch cables **2x**
Included with switch (AD)

STEP 28 Fit the battery and switch



Keep the battery disconnected and the switch OFF. Seat the battery in the large compartment. Push the switch into the side opening until it clicks into place.

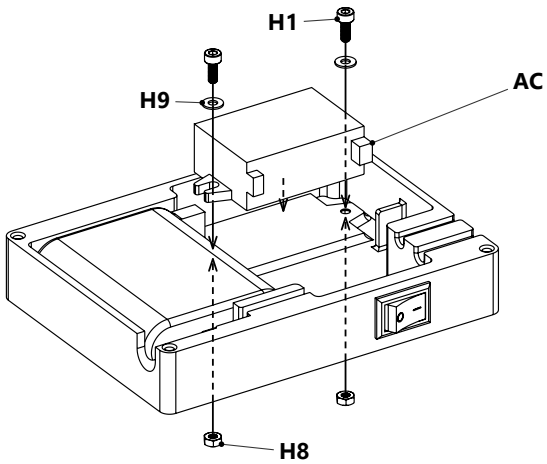
PARTS ADDED

P **1x**
Battery case bottom

AB **1x**
Battery

AD **1x**
Power switch

STEP 29 Mount the converter



Place the converter beside the battery. Secure it with two M3 × 8 mm screws and two washers. The two nuts go beneath the case.

PARTS ADDED

AC **1x**
DC-DC converter

H1 **2x**
M3 × 8 mm

H8 **2x**
M3 nut

H9 **2x**
M3 washer

ALREADY ASSEMBLED

Battery pack
From Step 28

STEP 30

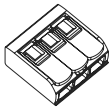
Wire the battery pack

Charging socket
Female barrel jack

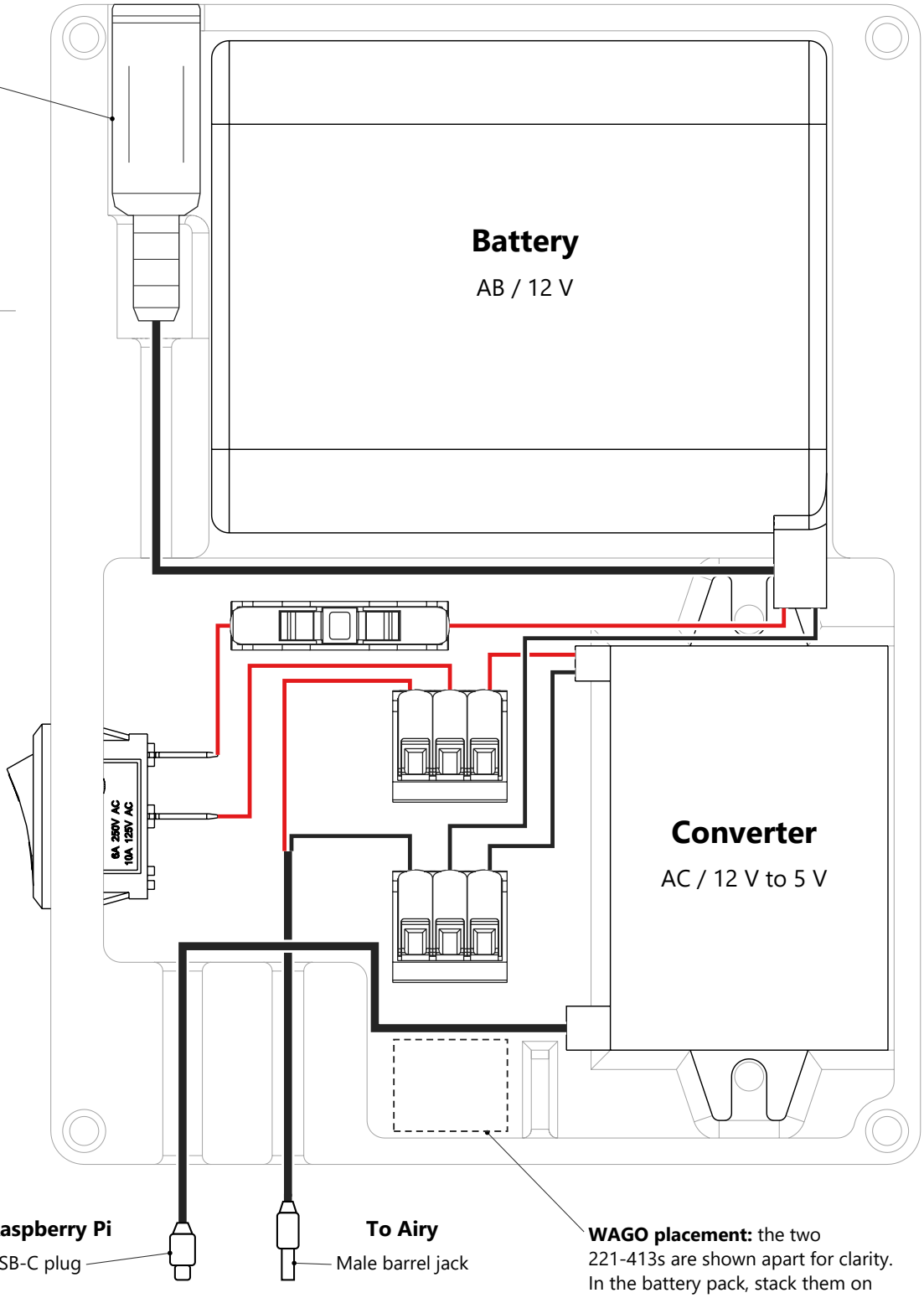
PARTS ADDED



AG **1x**
Inline WAGO
221-2401



AF **2x**
3-hole WAGO
221-413



Positive (+) Negative (-)

Using WAGO:

1. Make sure each wire is stripped to 11 mm.
2. Lift the lever fully, to 90 degrees.
3. Insert one wire until you can see it reach the opposite end through the clear walls.
4. Lower the lever, then tug the wire gently to check that it is clamped.

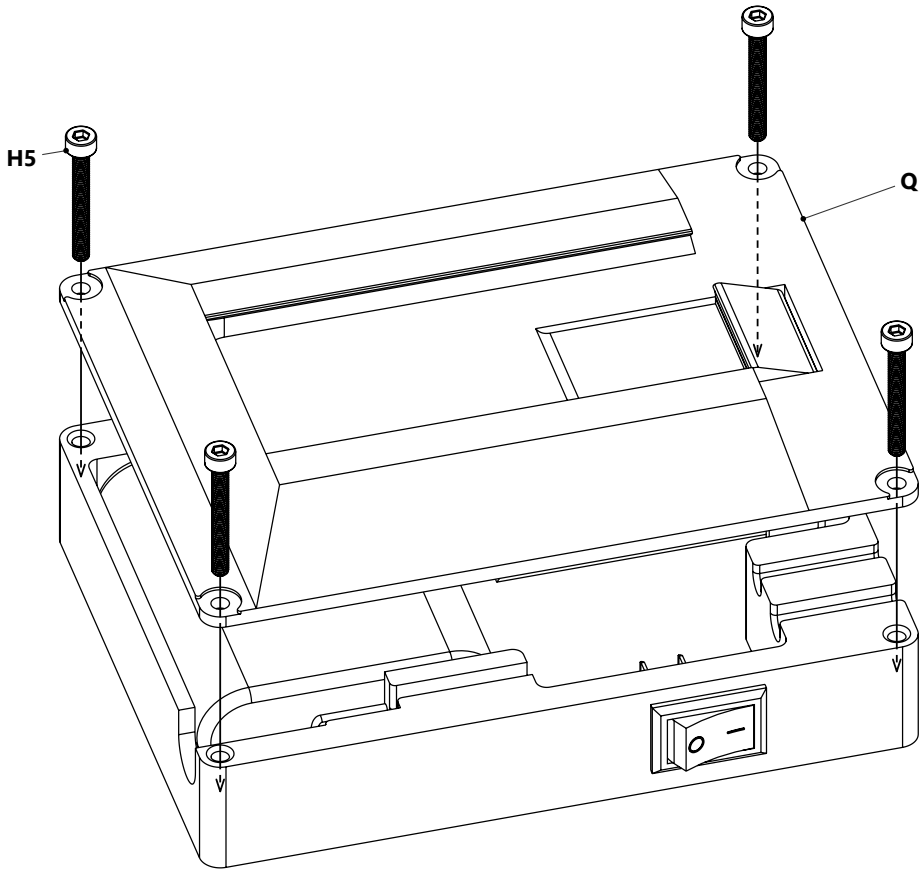
Output test: leave the Pi and Airy unplugged. Connect the battery and switch ON. Set the multimeter to DC volts: red probe in the center of the male barrel jack, black probe on the inside of its outer sleeve. Expect about +12 V. Then switch OFF and disconnect the battery.

STEP 31

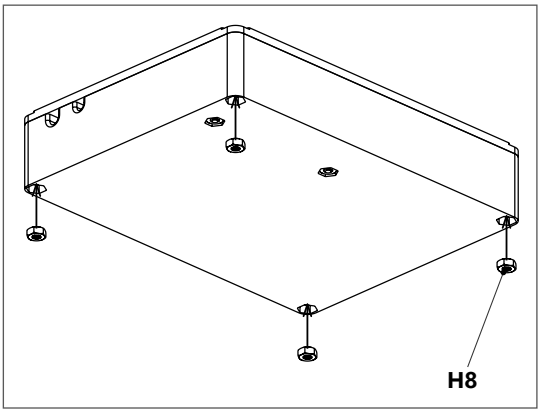
Close the battery pack

Finish the wiring and output test in Step 30 before closing the case.

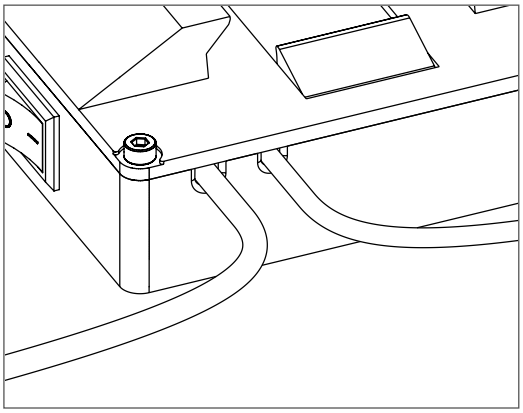
FIT THE COVER



UNDERSIDE



CABLE EXITS



1. After the output test passes, keep the switch OFF and reconnect the battery. Put each output cable into its case outlet.
2. Lower the lid carefully. Keep every wire away from the lid edge, screw posts, and screw holes.
3. Fit four M3 × 25 mm screws and four nuts. Check that the switch is accessible and both cables are held securely.

PARTS ADDED

Q **1x**

Battery case top

H5 **4x**

M3 × 25 mm

H8 **4x**

M3 nut

ALREADY ASSEMBLED

Wired battery pack

From Step 30

STEP 32

Install the battery pack

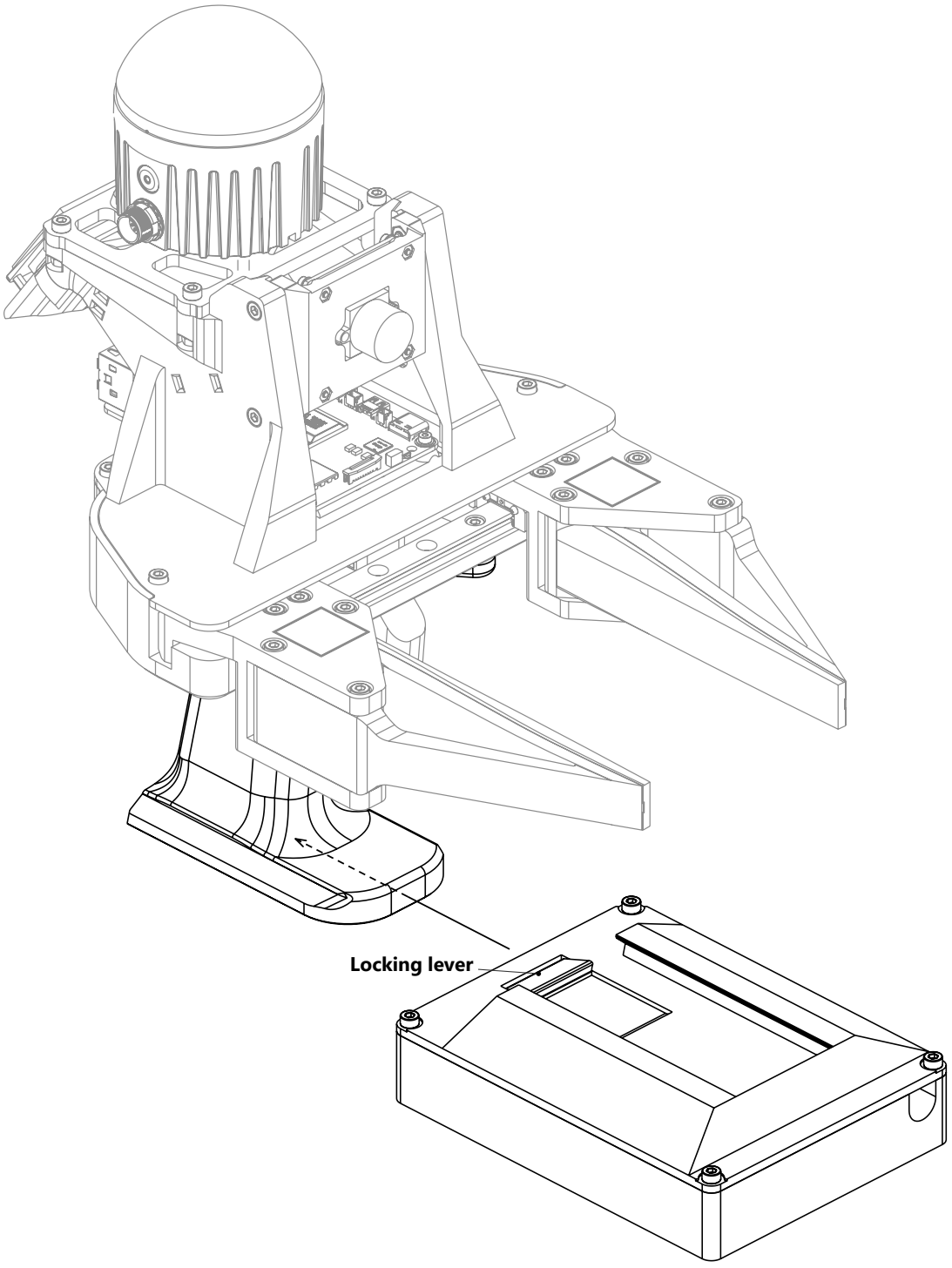
ALREADY ASSEMBLED

UMI-Live assembly

From Step 26

Battery pack

From Step 31

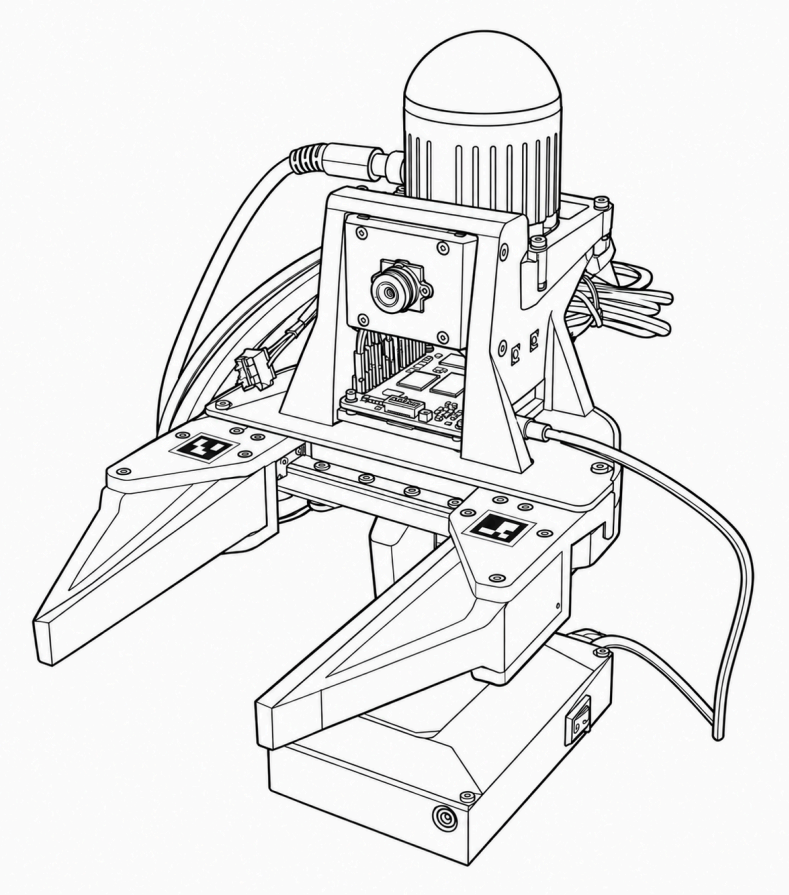


Align the battery pack rails with the base of the grip. Slide the battery pack until it clicks and locks into place.

If the locking lever cannot be pushed down, open the battery pack and make sure that the components inside are not blocking it.

STEP 33

Connect the cables



CABLE ROUTING REFERENCE

1. Hand-tighten the LiDAR cable onto the LiDAR connector.
2. Connect the Ethernet cable between the LiDAR cable and the Pi Ethernet port.
3. Connect the battery pack's male barrel jack to the LiDAR cable power input.
4. Connect the battery pack's USB-C output to the Pi power input.
5. Check the camera USB-A 3.0 connection, LCD cable, and button leads.
6. Optionally, tie the LiDAR cables together with zip ties to keep them organized.

PARTS ADDED

AK **1x**

Ethernet cable

1x

LiDAR cable

Included with LiDAR

Optional

Zip ties

ALREADY ASSEMBLED

Completed hardware

From Step 32

Final check

☐ Trigger springs back and both fingers move smoothly together.

☐ ArUco tags are flat and secured.

☐ The battery pack is locked in place and all cables are connected.

Charging the battery

1. Switch OFF. Disconnect all connections to the battery, including both bare-wire leads from the WAGO harness.
2. Plug the supplied charger directly into the battery's female barrel jack (charging socket).
3. Unplug the charger before reconnecting the battery to the harness.

Do not charge and use the battery at the same time.